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3 JULY 1963
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MILITARY SPECIFICATION

BINOCULAR, M13A1

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers 6 by 30 binoculars with a 30 millimeter objective aperture and an 8 degrees 30 minutes field of view.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-106 — Tape, Masking; Paper (Pressure - Sensitive).

UU-T-111 — Tape, Paper Gummed (Sealing and Securing).

PPP-B-621 — Boxes; Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes; Fiber.

PPP-T-76 — Tape, Pressure - Sensitive, Adhesive, Paper, Water Resistant.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior Packaging.

MIL-C-13459 — Cases, Binocular: M62A1 and M63A1.

MIL-O-13830 — Optical Components for Fire Control Instruments, General Specification Governing the Manufacture, Assembly and Inspection of.

MIL-F-13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-13988 — Paper, Lens, Tissue, Antitarnish, Wrapping.

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

FSC 6650

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MIL–I–45208 — Inspection Requirements, General Specification for.

MIL–I–45607 — Inspection Equipment, Supply and Maintenance for Ordnance.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

A7597721 — Binocular: M13A1.

IEL A7597721 — Index of Inspection Equipment Lists for Binocular M13A1.

LSQAP 22–382 — List of Supplementary Quality Assurance Provisions for Binocular M13A1.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitations for bids shall apply.

NATIONAL BUREAU OF STANDARDS

NBS Resolution Test Chart of 1952 (High Contrast) (Number 28X).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Construction. The binocular shall be manufactured in accordance with Drawing A7597721 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics to insure inherent quality and performance capabilities.

3.3 General specifications. The contractor shall be responsible for adherence to and compliance with following requirements of Specifications MIL–F–13926 and MIL–O–13830.

3.3.1 *MIL–F–13926.*

(a) Order of precedence.

(b) Dimensions and tolerances.

(c) Inorganic protective surface finishes.

(d) Part identification and marking.

(e) Workmanship.

3.3.2 *MIL–O–13830.* Cleanliness.

3.4 Environmental.

3.4.1 *Storage temperature.* The binocular shall show no evidence of glass breakage or other physical failure and shall meet all requirements of this specification at standard ambient temperature (60 to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of — 80° F. and + 160° F.

3.4.2 *Operating temperatures.* The binocular shall meet the applicable requirements of 3.10.3.1 and 3.11.1.1 while exposed and thermally stabilized at ambient temperatures of — 40° F. and + 150° F.

3.4.3 *Vibration.* There shall be no visible damage or loosening of assembled parts as a result of vibration in a vertical plane at a constant frequency of 30 cycles per second

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with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration, the binocular shall meet all of the following requirements of this specification.

3.5 Sealing. The binocular shall show no evidence of leakage when subjected to an external pressure of 5 psi gage pressure for a period of five minutes after having been exposed to the requirements of 3.4.

3.6 Resolution. The resolution of each monocular on the optical axis for a parallel bar test target shall be 6.5 seconds of arc or less.

3.7 Magnification. Magnification of each of the two telescopic systems shall be as specified on the drawings within a tolerance of 2 percent.

3.8 Collimation. When two parallel rays are projected into the two objectives respectively, the conjugate rays emerging from the two eyepieces shall be parallel within the following limits at the infinity focus of the eyepiece, and for any setting of the interpupillary distance scale between 58 and 72 mm as follows:

(a) There shall be no more than 15 minutes of arc in dipvergence.

(b) They shall not diverge more than 24 minutes of arc.

(c) There shall be no convergence.

3.8.1 *Image tilt and parallelism of images.* The images of a plumb line formed by the two optical systems shall be vertical within 1° and parallel within 0.5°.

3.9 Reticle. The horizontal reticle line shall be parallel to the line of centers of the right and left objective for a setting of 63 ± 1 millimeter interpupillary distance adjustment.

3.9.1 *Dioptric difference.* The eyepiece focus difference between the image of an infinity target and the reticle graduation nearest the center of the field shall not exceed ± 1/4 diopter.

3.9.2 *Reticle angle.* The reticle in the finished instrument shall indicate the correct angular values within 3 percent.

3.10 Eyepiece.

3.10.1 *Diopter scale.* Each diopter scale index shall indicate zero within a tolerance of ± 1/4 diopter when the image of an infinite target near the center of the field is brought into sharp focus with the aid of a dioptometer or an auxiliary telescope. Other graduations shall be accurate with ± 1/4 diopter.

3.10.2 *Diopter setting.* With the same diopter setting on each eyepiece, one eyepiece shall not extend beyond the other by more than 1/16 inch.

3.10.3 *Torque.* The torque required to adjust the eyepiece from — 4 to + 4 diopters shall be between 2.7 and 5.5 ounce inches when the eyepiece is rotated from — 4 to + 4 diopters in 5 seconds at standard ambient temperature (60° to 90° F.). The torque in any one eyepiece shall not vary more than 1.5 ounce inches.

3.10.3.1 *Operability.* The eyepiece shall be operable at the temperatures specified in 3.4.2.

3.11 Interpupillary adjustment. The interpupillary distance indicated by the scale shall be correct to within 1 mm.

3.11.1 *Torque.* The torque required to adjust the interpupillary setting between 58 and 72 mm shall be between 15 and 22 pound inches at standard ambient temperature (60° to 90° F.). The torque in any one instrument shall not vary more than 3 pound inches. The maximum torque required at the temperatures specified in 3.4.2 shall not exceed 43 pound inches.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Except where otherwise specified herein, the contractor's inspection system shall be in accordance with Specification MIL–I–45208.

4.3 Lot formation. When inspection lot sizes and lot formations are applicable, they shall be in accordance with Standard MIL–STD–105 and shall also be defined in the contractor's inspection system.

4.3.1 *Defective units and lots.* Defective units and lots shall be processed as specified in Standard MIL–STD–105 and shall not be included as part of the quantity specified in the contract until formal approval is obtained.

4.4 Inspection provisions. Components and subassemblies shall be examined in accordance with the inspection provisions contained in Specification MIL–F–13926 and 4.4.1 herein. Test and control of materials not covered by Supplementary Quality Assurance Provisions (SQAP's) or section 4 of Specification MIL–F–13926 shall be in accordance with "Test Data" provisions of Specification MIL–I–45208. Inspection related to section 3 of this specification shall be performed in accordance with 4.4.2. The tabulated Classification of Defects listed herein shall constitute the minimum inspection to be performed by the contractor prior to Government acceptance. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.4.1 *Inspection of components.* Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP) 22–382. Components not covered by SQAP's shall be inspected in conformance with Specification MIL–F–13926.

4.4.1.1 *Contents of SQAP.* The contents of a SQAP represent the minimum inspection requirements and shall be the basis for acceptance or rejection by a lot of each component covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual nonconforming items.

4.4.1.2 *Contractor's inspection for all components not covered by SQAP.* Inspection for these components shall be in accordance with the inspection procedures of the inspection system required by Specification MIL–I–45208 and the sampling plan tables of Specification MIL–F–13926. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the Binocular, M13A1. In addition, the list of class 1 defects in Specification MIL–F–13926 shall apply.

4.4.2 *Classification of defects and sampling plans.* Examination and testing to determine acceptance or rejection in compliance with the applicable requirements of section 3 shall be conducted on a 100 percent inspection basis, initially. After conclusive evidence of the process average has been established in conformance with Standard MIL–STD–105 and the AQL's specified herein, the classification of defects and sampling plans specified in tables I and II may be used in lieu of 100

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percent inspection. The AQL value specified shall apply to each individual characteristic on a single defect basis.

4.4.2.1 *Disposition of rejected lots.* Rejected lots shall be screened in conformance with 4.3.1 for all defective characteristics causing rejection of the lot. After the Government inspector has approved the corrective action taken, subsequent production lots may be submitted for acceptance in accordance with 4.4.2 and Standard MIL–STD–105.

TABLE I. *Classification of defects*

Use Inspection Level II, in Table IIIA with Sampling Plan Table IVA of Standard MIL–STD–105.

Critical: None defined.

Major	AQL 1.0 percent defective	Requirement paragraph	Quality assurance paragraph	Test method
101	Physical failure, post vibration	3.4.3	4.6.3	Visual
102	Insufficient resolution	3.6	4.6.5	Standard measuring equipment (SME).
103	Incorrect magnification	3.7	4.6.6	SME.
104	Improper collimation	3.8	4.6.7	SME.
105	Excessive divergence	3.8	4.6.7.1	SME.
106	Excessive divergence-convergence	3.8	4.6.7.2	SME.
107	Incorrect image tilt and parallelism	3.8	4.6.7.3	SME.
108	Reticle plumb not within limits	3.9	4.6.8	SME.
109	Dioptric difference not within limits	3.9.1	4.6.8.1	SME.
110	Incorrect reticle angle	3.9.2	4.6.8.2	SME.
111	Incorrect diopter scale readings	3.10.1	4.6.9	SME.
112	Improper diopter setting	3.10.2	4.6.10	SME.
113	Eyepiece torque not within limits	3.10.3	4.6.11	SME.
114	Incorrect interpupillary adjustment	3.11	4.6.12	SME.
115	Interpupillary torque not within limits	3.11.1	4.6.12.1	SME.

Minor: None defined.

Note. The above tests shall be conducted at standard ambient temperature (plus 60 to plus 90 degrees Fahrenheit) in accordance with the Quality Assurance Paragraph (Q.A. par.) listed.

TABLE II. *Classification of defects*

Use Inspection Level L7, in Table IIIB with Sampling Plan Table IVA of Standard MIL–STD–105.

Critical: None defined.

Major	AQL 1.0 percent defective	Requirement paragraph	Quality assurance paragraph	Test method
101	Sealing failure	3.5	4.6.4	SME.
102	Operability failure	3.10.3.1	4.6.11.1	SME.
103	Excessive interpupillary torque	3.11.1	4.6.12.1	SME.

Minor: None defined.

Note. The above tests shall be conducted at ambient temperatures of minus 40 degrees and plus 125 degrees Fahrenheit in accordance with the Quality Assurance Paragraph (Q.A. par.) Listed, except the sealing test which shall be conducted at standard ambient temperature, subsequent to the operating test.

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4.4.3 *Control inspection.* The first 5 binoculars produced, and 3 selected at random from every 100 subsequently produced (or from each month's production, whichever occurs first) shall represent the control samples for control inspection. Control samples shall be subjected to the examinations and tests specified herein to determine compliance with 3.1, 3.2, 3.3 (b) (c) (d) (e) and 3.4. The examinations and tests shall be conducted by the contractor under surveillance of the Government representative at the contractor's facility, or Government approved laboratory, using test equipment authorized for use by the contractor, or otherwise approved by the contracting officer.

4.4.3.1 *Control sample failure.* Should a control sample fail to meet the requirements specified in 4.4.3, the contractor shall cease those operations causing rejection until necessary corrections have been made by the contractor and approved by the Government. When production has resumed, inspection of those requirements that have previously caused rejection shall be performed on a 100 percent basis until 10 consecutively produced binoculars meet these requirements. When this is accomplished, control inspection (4.4.3) may be continued.

4.5 Inspection equipment. Inspection equipment shall conform to drawings listed in IEL A7566960 except as otherwise provided for under the alternative inspection procedure and inspection equipment requirements of Specification MIL–I–45208. Provisions of Specification MIL–I–45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

4.5.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except in 4.5.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exhorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.5.2 *Inspection equipment design requirements.*

4.5.2.1 *Device to test temperature requirements.* The device to test temperature requirements shall meet the inspection requirements of Specification MIL–F–13926.

4.6 Test methods and procedures.

4.6.1 *Storage temperature.* This test shall be performed in accordance with Procedure II of Specification MIL–F–13926 in a chamber capable of maintaining the temperatures specified. Upon completion of the specified exposure and return to standard ambient temperature (60° to 90° F.) the binoculars shall be inspected for damage in compliance with 3.4.1 and then subjected to all the tests of this specification.

4.6.2 *Operating temperature.* This test shall be conducted in accordance with Procedure I of Specification MIL–F–13926 except that the temperature shall not be reduced beyond condition "C" and not raised beyond condition "G." When the binocular is thermally stabilized at "C" and "G," in compliance with 3.4.2, it shall be subjected to the tests specified in 4.6.11.1 and 4.6.12.1 and shall meet the requirements specified.

4.6.3 *Vibration.* The binocular shall be mounted on a suitable adapter and vibrated by means of a vibration test fixture capable of vibrating the binocular in a vertical plane at the frequency, amplitude and duration as indicated in 3.4.3. After vibration, the binocular shall be visually inspected for damage,

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and shall then be subjected to all the tests of this specification and shall meet the requirements specified.

4.6.4 *Sealing.* The binocular shall be placed in a suitable container connected to a pressure test fixture which shall include a sensitive pressure gage capable of testing the requirements of 3.5. The container, with the binocular, shall then be charged with 5 psi gage pressure, stabilized, and the source cut off. Evidence of leakage can be determined by observing the gage for pressure drop for the time interval specified in 3.5.

4.6.5 *Resolution.* Resolution test is accomplished with the binocular positioned as specified in the collimation test of 4.6.7. The "56" pattern of the National Bureau of Standards Test Chart 25X is attached to the central area of the wall chart located 46 feet from the objectives of the binocular. The test chart shall be illuminated by a (60) watt light located one foot from the chart. With the telescope focused on the "56" numeral, the test patterns on either side of the "56" numeral are observed through a three-power dioptometer or equivalent placed in front of each eyepiece of the binocular. The test pattern shall be resolved within the arc specified in 3.6. The test pattern shall be considered resolved when the observer can determine the direction in which the lines lay.

4.6.6 *Magnification.* The magnification test shall be in accordance with Specification MIL–O–13830 for the requirements of 3.7.

4.6.7 *Collimation.* The test for collimation is made by means of a special testing device for collimation. The device shall consist of three collimating telescopes with reticles for observing wall or collimator targets through the binocular at the 58 and 72 mm interpupillary positions. A mounting support shall be designed to position the binoculars set at 72 mm so that the line of sight of two telescopes will pass through the line of sight of the binoculars and focus on two of the wall targets or collimators. The third telescope will be located so the line of sight will pass through the right eyepiece and focus on the third target when the binocular is set at 58 mm. With the binocular set at the seventy-two (72) mm interpupillary distance, it is clamped in the holder of the device and positioned to coincide the horizontal line of the binocular (in the left eyepiece) with the lower horizontal line of the target at the zero mark. In this position, any error in collimation can be read on the right target of the device at both the 58 mm and 72 mm interpupillary distances. The binocular shall be checked in accordance with the following test procedures to determine compliance with the requirements of 3.8.

4.6.7.1 *Dipvergence.* Dipvergence is indicated by the comparison of vertical displacement of the target on the grid of the collimator as viewed at both 58 mm and 72 mm interpupillary distances, and shall not exceed the requirement of 3.8.

4.6.7.2 *Divergence and convergence.* Divergence is indicated when the target appears in the right half of the grid and convergence when in the left half as viewed in the right hand collimating telescopes. Divergence shall not exceed the tolerance specified in 3.8. There shall be no convergence.

4.6.7.3 *Image plumb.* The images are checked for plumb and parallelism through the two optical systems by means of the vertical lines of the collimating telescope and the vertical lines of the target. A tolerance block on the target will be used to determine conformance with the requirements of 3.8.

4.6.8 *Reticle.* With the binocular set at 63 mm interpupillary distance and mounted on a fixture with the line of centers parallel to the target horizontal line, the horizontal reticle line shall coincide with the target horizontal line within the tolerance of 3.9.

4.6.8.1 *Dioptric difference.* The dioptric difference shall be determined with the aid of a dioptometer of at least three (3X) power.

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With the line of sight directed at an infinity target, the eyepiece of each monocular shall be adjusted for sharp focus and shall not exceed the tolerance specified in 3.9.1.

4.6.8.2 *Reticle angle.* The angular subtense of the reticle markings shall be measured with the aid of a calibrated target and shall be within the limits specified in 3.9.2.

4.6.9 *Diopter scale.* The eyepiece focus adjustment shall be performed with the aid of a dioptometer of at least three (3X) power. The image of an infinity target shall be brought into sharp focus on each eyepiece and the zero and other readings shall coincide within the tolerance of 3.10.1.

4.6.10 *Diopter setting.* The diopter scale of each monocular shall be adjusted to the same settings. A straight edge is then placed across the rings of both eyepieces and neither shall extend beyond the other more than the tolerance specified in 3.10.2.

4.6.11 *Eyepiece torque.* The torque required to rotate each eyepiece shall be measured with a torque wrench with an adapter and shall be within the limits specified in 3.10.3.

4.6.11.1 *Operability.* While the binocular is exposed to the extreme operating temperatures specified in 3.4.2, the eyepieces shall be rotated to insure correct operation as specified in 3.10.3.1.

4.6.12 *Interpupillary adjustment.* The interpupillary setting shall be measured by means of a metric scale between the inside and outside edges of the eyepieces and this measurement shall coincide with the scale of the binocular within the tolerance of 3.11.

4.6.12.1 *Torque.* The torque required to adjust the interpupillary setting within the limits specified in 3.11.1 shall be measured by means of a standard measuring equipment at the specified temperatures. With one side of the binocular held rigid, a direct pull is applied to the outer side for movement within the torque limits specified in 3.11.1.

4.6.13. *Preservation, packaging, packing and marking.* Quality assurance provisions for preservation, packaging, packing and marking shall be in accordance with the applicable specifications and the prescribed method of unit protection indicated in Specification MIL–P–14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with figs. 1 through 4.

6. NOTES

6.1 Intended use. The binocular covered by this specification is used for observation and approximate measurement of small angles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable stock number.
(c) Selection of applicable levels of packaging and packing required.

Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—MU
Navy—BuShips
Air Force—MAA

Preparing activity:
Army—MU
Project No. 6650–0092

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FEDERAL ITEM NAME		PART NO.	FEDERAL STOCK NO.
BINOCULAR: M13A1 W/E		6702509	6650-670-0973
COMPOSITION Plastic, Glass, Aluminum Rubber and Steel	FINISH Black coated surfaces and paint		

LOGISTICS DATA	LEVEL A MILITARY				LEVEL B LIMITED MILITARY			
	QTY PER PACK	EXTERIOR SIZE	GR WT	CUBE	QTY PER PACK	EXTERIOR SIZE	GR WT	CUBE
ITEM		7½ x 6⅛ x 4⅛	3.23	.110		SAME AS LEVEL A		
UNIT PACKAGE	1	8¼ x 7⅝ x 4⅝	4.14	.178		SAME AS LEVEL A		
INTERMEDIATE PKG.		NONE				NONE		
EXTERIOR PACK	24	34 x 26¼ x 11	140.	5.62				

PACKAGING REQUIREMENTS	MIL-P-116 UNIT PROTECTION: METHOD IA8 CLEANING (A) DRYING (A)						MIL-P-116 UNIT PROTECTION: METHOD CLEANING DRYING					
	DWG. OR SPEC.	SIZE	STYLE	TYPE	GRADE	CLASS	DWG. OR SPEC.	SIZE	STYLE	TYPE	GRADE	CLASS
TISSUE 3	MIL-P-13988	126 sq. in. (B) *						SAME				
TAPE 4,5,6	UU-T-106	39 x 1 (B)(C)(D)		I-II								
CUSHIONING 7	PPP-P-291	21 x 12 (E)		I		1		AS				
TAPE 8	UU-T-111	15 x 2 (E)				2						
BAG 9	MIL-B-117	10½ x 10¼ (F)		I		e		LEVEL				
CONTAINER 10	PPP-B-636	8 x 7½ x 4¼	RSC		W6c	2						
CLOSURE 11	PPP-T-76	27 x 2 (G)						A				
	INTERMEDIATE PACKAGE						INTERMEDIATE PACKAGE					
	DWG. OR SPEC.	SIZE	STYLE	TYPE	GRADE	CLASS	DWG. OR SPEC.	SIZE	STYLE	TYPE	GRADE	CLASS
CONTAINER												
CLOSURE		NONE						NONE				

PACKING REQUIREMENTS	EXTERIOR PACK TYPE OF LOAD 2 JAN-P-100						EXTERIOR PACK TYPE OF LOAD 2 JAN-P-100					
	DWG. OR SPEC.	INSIDE DIMENSIONS	STYLE	TYPE	GRADE	CLASS	DWG. OR SPEC.	INSIDE DIMENSIONS	STYLE	TYPE	GRADE	CLASS
CASE LINER												
CONTAINER	PPP-B-621	31 x 24⅜ x 9½ (H)	4			2		SAME AS				
CLOSURE		(J)										
								LEVEL C				
STRAPPING	QQ-S-781-790	SEE APPLICABLE BOX SPEC					QQ-S-781-790	SEE APPLICABLE BOX SPEC				
MEASUREMENTS SIZES IN INCHES WEIGHT IN LBS. CUBAGE IN FT.	LEVEL C MINIMUM MILITARY PACKAGE AS SPECIFIED IN MIL-P-14232						LEVEL C MINIMUM MILITARY PACK AS SPECIFIED IN MIL-P-14232					

MARKING - *(Packages and Containers shall be marked as specified by MIL-STD-129 by referenced specifications or notes below).*

NOTES: * See alternate method on Figure 3

See Figures 2 and 3 for explanation of Notes (A) through (J) and Marking Notes.

The numbers in the packaging requirements column indicate the sequence of packaging operations.

FIGURE 1

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FEDERAL ITEM NAME		PART NO.	FEDERAL STOCK NO.
BINOCULAR: M13A1 W/E		6702509	6650-670-0973
COMPOSITION Plastic, Glass, Aluminum, Rubber and Steel	FINISH Black coated surfaces and paint		

ITEM	PART NO.	FEDERAL STOCK NUMBER	WEIGHT	PACKAGED WT.
Binocular, M13A1 W/E	6702509	6650-670-0973	3.15	4.06
Case, Carrying M62A1 (MIL-C-13459A(MU))	7694280	6650-769-4280	1.60	

NOTES:

(A) Optics shall be cleaned and dried as specified in MIL-P-14232. Process C-1 of MIL-P-116 shall be followed to clean other surfaces. Except for optics, surfaces shall be dried by any procedure specified in MIL-P-116 that will not damage the items.

(B) Two pieces of tissue (9 x 3) shall be folded to form 2 pads 3 x 2¼. A pad shall be placed over each eyepiece. The tissue over the eyepieces shall then be secured in place with tape (2 pcs, ea 5 x 1). Two pieces of tissue (12 x 4) shall be folded to form 2 pads 4 x 3. A pad shall be placed over each objective end of the binocular. The tissue shall then be secured in place with tape (2 pcs., ea 7 x 1).

(C) The shoulder straps on the case shall be folded and secured to the case with tape (4 pcs., ea. 3 x 1) as shown in Figure 4.

(D) The binocular shall be placed in the case. The neck strap shall be coiled and tape (3 x 1) shall be applied over the coil as shown in Figure 4 to prevent it from uncoiling. The coiled strap shall be placed between the two objective ends of the binocular as shown in Figure 4. The lid shall be closed and secured with the latch provided on the case.

(E) The carrying case containing the binocular shall be wrapped with paperboard cushioning (21 x 12) in such a manner that two thicknesses of paperboard cushioning is over the latch. The wrap shall be secured in place with tape (3 pcs., ea. 5 x 2).

(F) The bag shall be heat sealed, exhausting entrapped air prior to final closure.

(G) Tape shall be applied over the top and bottom seams and extend at least 3 inches onto each end panel. If the bottom of the container is closed with staples or adhesive, only a single strip of tape shall be applied over the top seam.

(H) Only binoculars with the same stock number shall be packed in the shipping container.

(J) The container shall be closed as specified in PPP-B-621.

FIGURE 2

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FEDERAL ITEM NAME		PART NO.	FEDERAL STOCK NO.
BINOCULAR: M13A1 W/E		6702509	6650-670-0973
COMPOSITION Plastic, Glass, Aluminum, Rubber and Steel	FINISH Black coated surfaces and paint		

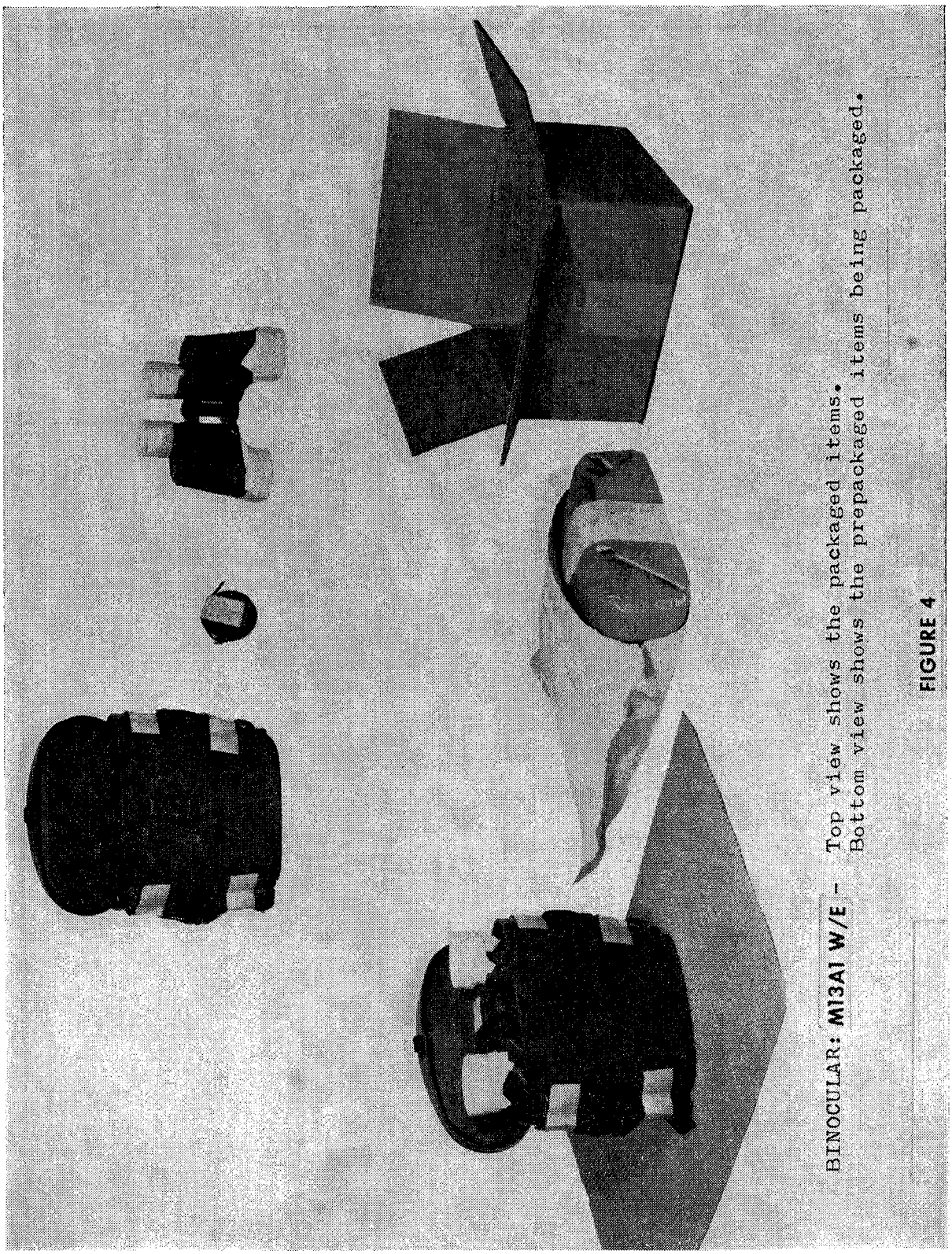
MARKING NOTES:

The serial number shall be included in the identification data for Level A packaging. Serial numbers of the binoculars in a Level A or B shipping container shall be included in the identification data. The nomenclature shall be omitted on all shipping containers.

ALTERNATE METHOD: Plastic caps conforming to type III, form A, class I of Specification MIL-C-52078 may be used to cover the eye-pieces and objective ends of the binocular in lieu of tissue paper and taping.

FIGURE 3

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BINOCULAR: M13A1 W/E - Top view shows the packaged items. Bottom view shows the prepackaged items being packaged.

FIGURE 4

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